

Grégoire Dhimoïla

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🔗 Parabrele

🎓 Google Scholar

Research Interests

Researcher in mechanistic interpretability and multimodal representation learning, with work on large language models, state space models and computational neuroscience.

Research Experience

2025 – 2026

Brown University — Serre Lab

Visiting Research Scholar, advised by Thomas Serre.

- First author on two NeurIPS 2026 submissions and coauthor on one additional submission on multimodal representations and state space models [1,2,3]
- Studied links between mechanistic interpretability and computational neuroscience
- Developed frameworks for analyses of representation learning in multimodal models
- Derived spectral formulations enabling parallel recurrent vision architectures.

Summer 2025

DEEL / IRT Saint Exupéry

Research Intern, advised by Thomas Fel and Agustin Picard.

- First-author paper accepted at ICLR 2026 [4].
- Introduced sparse decomposition methods separating shared and modality-specific embedding geometry
- Enabled applications like semantic vector arithmetic and modality-gap-free embeddings.

2024 – 2025

Cadenza Labs

Research Scholar, advised by Walter Laurito.

- Co-first author paper accepted at EMNLP 2024 [5]
- Developed Contrast-Pair Clustering methods for CCS-style approaches and other mechanistic interpretability projects.

Summer 2024

Goethe University Frankfurt

Research Intern, advised by Holger Dell.

- Developed computational graph structures based on aggregates of dictionary atoms
- Investigated unsupervised behavioral control in neural networks through systematic interventions on the computational graph.

Education

2022 – 2026

ENS Paris-Saclay

Master 2 Mathematics, Vision, Learning (2024 – 2025)

Master 1 Parisian Master of Research in Computer Science (2023 – 2024)

L3 Mathematics and Computer Science (2022 – 2023)

2020 – 2022

Lycée Aux Lazaristes

Classe préparatoire MP* — intensive two year course preparing for the highly competitive national entrance examination to the French Grandes Écoles.

Publications & Preprints

Conference Proceedings

- [4] | **Grégoire Dhimoïla**, Thomas Fel, Victor Boutin, and Agustin Martin Picard
Cross-modal redundancy and the geometry of vision-language embeddings
ICLR 2026
- [5] | Walter Laurito*, Sharan Maiya*, **Grégoire Dhimoïla***, Owen Ho Wan Yeung, Kaarel Hänni
Cluster-Norm for Unsupervised Probing of Knowledge
EMNLP 2024

Preprints / Under Review

- [1] | **Grégoire Dhimoïla**, Victor Boutin, Agustin Martin Picard, Thomas Fel, and Thomas Serre
A Unifying Framework for Concept-Based Representational Similarity
NeurIPS 2026 submission
- [2] | Drew Linsley*, **Grégoire Dhimoïla***, Alekh Karkada Ashok, Brian Kim, and Thomas Serre
Recurrent Vision in Parallel: Spectral Convolutional State Space Models
NeurIPS 2026 submission
- [3] | Yu-Ang Cheng, Sixuan Chen, Zhouyang Lu, Xizheng Yu, **Grégoire Dhimoïla**, Thomas Serre
A Unified Spectral Theory of Multimodal Losses
NeurIPS 2026 submission